PLAN

A web-based AI-powered examination management, question generation, and learner assessment platform, designed around Uganda's National Curriculum Development Centre (NCDC) curriculum and Uganda National Examinations Board (UNEB) assessment standards

ACTORS;

1. Teachers: Generate examinations, create question banks, mark scripts, and analyse learner performance.
2. Schools: Manage examinations, classes, subjects, results, and academic reports.
3. Learners: Take examinations, receive feedback, practise questions, and monitor their academic progress.

Expected functionalities;

1. AI examination question generator

Teachers select class, subject, topic, learning outcome, difficulty, question type, and number of questions. AI generates curriculum-aligned questions.

2. Curriculum-based question bank

Stores approved questions, answers, marking guides, difficulty levels, topics, and learning outcomes. Teachers can edit and reuse questions.

3. Online examination system

Teachers publish examinations. Learners log in, answer questions, save responses, and submit their examinations.

4. AI-assisted marking

Automatically marks objective questions and assists teachers with marking structured, essay, and open-ended questions using approved marking guides.

5. Performance analytics

Identifies weak topics, class performance, individual learner progress, difficult questions, and areas requiring remedial teaching.

6. Report generation

Generates report cards, examination reports, class rankings, subject performance summaries, and downloadable PDF/Excel reports.

7. AI learning assistant

Explain answers, provides hints, recommends revision materials, and generates additional practice questions based on learner weaknesses.

8. School administration

Manages schools, teachers, learners, classes, academic years, subjects, examination schedules, and user permissions.

The primary source should be the National Curriculum Development Centre (NCDC). Its official resources include primary, lower secondary, advanced secondary, and assessment materials.

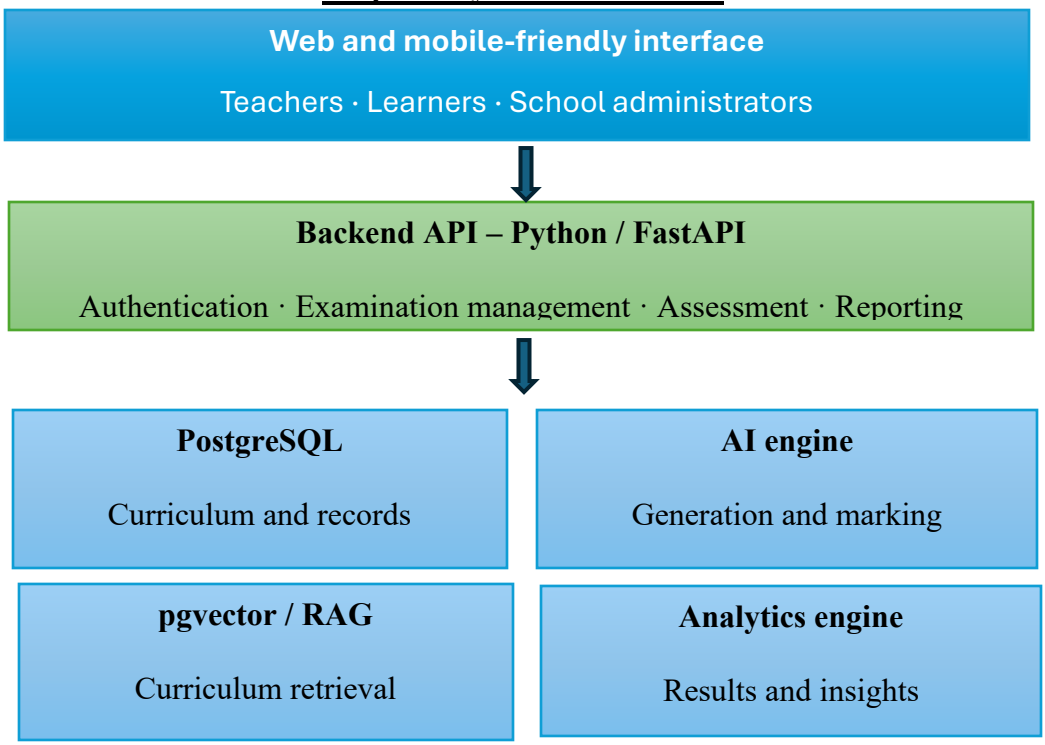
Education level	**Classes**
Primary	P1–P7
Lower secondary	S1–S4
Advanced secondary	S5–S6

*****_The NCDC Lower Secondary Curriculum Framework explicitly emphasizes knowledge, understanding, skills, values, and learner competencies. Your question generator must account for these dimensions rather than simply generating recall questions._

Technology and infrastructure

Component	**Recommended technology**	**Purpose**
Frontend	React / Next.js	User interfaces
Backend	Python / FastAPI	Application logic and APIs
Database	PostgreSQL	Schools, curriculum, questions, results
AI engine	OpenAI API	Question generation and marking assistance
Vector database	pgvector	Curriculum retrieval and semantic search
File storage	S3-compatible storage	Documents, scripts, attachments
Background jobs	Celery + Redis	Large AI jobs and report processing
Analytics	Python, Pandas	Performance analysis
PDF generation	ReportLab	Examination papers and reports
Hosting	Cloud VPS or managed cloud	Production deployment

Proposed system architecture



The workflow example:

1. Teacher selects Senior Two Mathematics.
2. Teacher selects Algebra and linear equations.
3. System retrieves the relevant NCDC learning outcomes and assessment guidance.
4. AI receives the curriculum extracts and examination instructions.
5. AI generates questions, model answers, marking guides, and difficulty classifications.
6. An automated validation process checks the output.
7. Teacher reviews and approves the examination before publication.

How the AI question generator is to be designed;

Question type	Example
Multiple choice	Select the correct answer
True/False	Determine whether a statement is correct
Short answer	State, define, or explain

Structured	Answer several related questions
Essay	Discuss, evaluate, or analyse
Practical	Perform an experiment or demonstrate a skill
Case study	Interpret information and solve a real-world problem

*A teacher specifies the level of difficulty of the questions.

AI-assisted marking and assessment

Automatic marking;

For; multiple-choice questions, True/False questions, matching questions., numerical questions with defined answers., fill-in-the-blank questions, these should be marked using deterministic rules rather than an AI model wherever possible.

AI-assisted marking

For: essays. explanations. structured questions. comprehension. questions with multiple acceptable answers, the AI compares the learner's response against an approved marking guide and suggests a score.

Practical assessment

This is especially important for competency-based education.

Teachers should be able to assess: Practical demonstrations, Projects, Group assignments, Presentations, Laboratory activities, Fieldwork.

The platform should support configurable rubrics and teacher-entered observations.

Important features

- Mobile-first design
- Low-bandwidth support
- Offline examinations
- Flexible assessment models
- Ugandan context

DATA PROTECTION AND SECURITY

⇒ Appropriate parental/guardian consent processes.
⇒ Role-based access control.
⇒ Encrypted communication using HTTPS.
⇒ Secure password storage.
⇒ Database backups.

⇒ Audit trails.
⇒ Data retention and deletion policies.
⇒ Restricted access to learner results.
⇒ Appropriate controls over AI providers receive learner information.
⇒ A privacy policy and terms of service.